AMERICAN ACADEMY OF OPHTHALMOLOGY®

Assessment of Early Fuchs Endothelial Corneal Dystrophy and CTG Trinucleotide Expansion Positivity Using Scheimpflug Imaging

Stanley S.J. Poh, MBBS,[1,2] *Kai Yuan Tey, MBBS,*[1] *Gary S.L. Peh, PhD,*[2,3] *Dawn J.H. Neo, MSc,*[3] *Hla Myint Htoon, PhD,*[4] *Evan K.L. Lee,*[4] *Yu Qiang Soh, MBBS,*[5] *V Vinod Mootha, MD,*[6] *Jodhbir S. Mehta, PhD*[1,2,3,4]

Purpose: To evaluate changes in corneal densitometry and optical aberrations using Scheimpflug imaging in early Fuchs endothelial corneal dystrophy (FECD) and to assess their association with CTG trinucleotide repeat expansion.

Design: Retrospective cross-sectional study.

Subjects: Fuchs endothelial corneal dystrophy eyes diagnosed between 2018 and 2022 were included. Control eyes were recruited from healthy individuals undergoing cataract surgery.

Methods: All eyes underwent Scheimpflug imaging. Subclinical edema was defined by the presence of ≥2 of the following features: (1) loss of parallel isopachs; (2) displacement of the thinnest point; and (3) focal posterior depression. A subset of FECD subjects was genotyped for CTG repeat expansion.

Main Outcome Measures: Corneal densitometry, higher-order aberration (HOA), and pachymetry were quantified. Densitometry was measured across 3 layers (anterior 120 μm, central, and posterior 60 μm) and within 4 concentric annuli while HOA was analyzed separately for the anterior and posterior cornea.

Results: One hundred eight FECD cases and 59 controls were included. Eyes with subclinical edema exhibited significantly higher densitometry and HOA across all corneal layers and zones compared with those without edema (all $P < 0.01$). Subclinical edema was independently associated with increased posterior corneal HOA ($\beta = 15.068$, 95% confidence interval [CI]: 7.546–22.590) and elevated densitometry in the central 0 to 2 mm annulus across all layers. Among genotyped eyes, 22 (44.0%) were positive for CTG expansion, which was associated with lower densitometry in the anterior ($\beta = -4.59$, 95% CI: −7.28 to −1.92), central ($\beta = -1.76$, 95% CI: −2.79 to −0.74), and posterior ($\beta = -3.05$, 95% CI: −5.67 to −0.44) layers of the central 0 to 2 mm annulus.

Conclusions: Fuchs endothelial corneal dystrophy eyes with subclinical edema are associated with increased corneal densitometry and HOA, indicating early structural and optical changes. In contrast, the presence of CTG repeat expansion was correlated with reduced central corneal densitometry, suggesting a potentially distinct disease phenotype.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100830* *© 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

Fuchs endothelial corneal dystrophy (FECD) is a progressive corneal disease characterized by the dysfunction of endothelial cells and the formation of endothelial excrescences termed guttae. These changes often lead to corneal edema in the later stages, significantly impacting vision.[1,2] The management of FECD has also evolved considerably in recent years.[3] With the increasing popularity of Descemet membrane endothelial keratoplasty, there is a growing trend toward earlier surgical intervention to prevent the formation of subepithelial or stromal scarring, thereby optimizing visual outcomes.[4,5] Identifying patients with early FECD who exhibit signs of early decompensation, such as subclinical edema and associated changes in densitometry or optical aberrations, is important because these individuals may benefit from timely surgical intervention.

The conventional Krachmer grading of FECD relies on slit lamp biomicroscopy to assess number and confluency of guttae, with corneal edema signifying the most advanced stage of disease.[6] However, this grading is highly variable between observers.[7,8] Alternative parameters such as central corneal thickness or central-to-peripheral thickness

© 2025 by the American Academy of Ophthalmology
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/). Published by Elsevier Inc.

https://doi.org/10.1016/j.xops.2025.100830
ISSN 2666-9145/25

ratio have been proposed as objective measures, but they are not sensitive in the early stages of disease.[7,9] In recent years, Scheimpflug imaging is increasingly adopted as an imaging modality of choice in FECD. The presence of subclinical edema tomographically, defined by loss of parallel isopachs, displacement of the thinnest point of the cornea, or focal posterior corneal surface depression, was found to have prognostic value in FECD.[10,11]

Densitometry provides a quantitative assessment of corneal optical clarity and has been applied in the evaluation of corneal opacity, corneal haze postrefractive surgery, and postoperative corneal edema.[12,13] In FECD, average densitometry, especially of the central cornea, is significantly elevated compared with controls and has been proposed as a surrogate marker for visual function.[14–17] Optical aberrations, another measure of optical clarity, can also be quantified using Scheimpflug imaging. Studies have shown that both anterior and posterior higher-order aberration (HOA) are increased in the early FECD compared with controls, reflecting a decline in visual quality attributable to microstructural changes, even in the absence of clinically apparent edema.[15] Together, densitometry and HOA represent objective metrics that may be particularly valuable for assessing visual function in the early stages of FECD.

Recent studies have identified a genetic association between FECD and trinucleotide repeat expansion at the intronic CTG18.1 locus of the TCF4 gene. The prevalence of CTG expansion (≥40 repeats) is notably higher in Western populations, reported at 50% in Australia, 69% in the United States, and 79% in Germany.[18–20] In contrast, the prevalence in Asian populations is lower, with 43.9% among Chinese individuals in Singapore, and 23% to 26% in Japan.[21–23] Patients with CTG expansion were shown to carry a twofold increased risk of disease progression.[21,24] Although CTG expansion is considered a potential causative factor and a marker of more severe disease, no definitive correlation has yet been established between repeat length and severity markers on Scheimpflug imaging.[23,25]

Given the versatility of Scheimpflug imaging in the detection of early morphologic changes, this study aimed to investigate the association between imaging-derived parameters and the presence of subclinical edema in FECD. We hypothesized that structural alterations, particularly involving the posterior cornea, contribute to increased densitometry and HOA even before the onset of clinically evident edema. Additionally, we also aimed to determine whether any of these imaging parameters could serve as predictors of CTG trinucleotide repeat expansion in the TCF4 gene.

Methods

This is a retrospective cohort study of participants with FECD who were recruited between 2018 and 2022. Fuchs endothelial corneal dystrophy was diagnosed based on the presence of central or paracentral confluent guttae on slit lamp examination. Patients with previous keratoplasty, significant corneal edema or scarring, poor image centration, and poor image quality on Scheimpflug imaging were excluded from this study. A separate cohort of controls was recruited from healthy participants scheduled for cataract surgery. This study was conducted according to the tenets of the Declaration of Helsinki and was granted approval by the SingHealth Centralized Institutional Review Board, protocol number R1509/8/2018. Due to the retrospective design and use of anonymized data, the requirement for patient consent was waived.

Image Acquisition

All participants underwent pupillary dilation with 1% tropicamide and 2.5% phenylephrine to both eyes. Sclerotic scatter and retroillumination photographs were captured on a slit lamp (SL-D8Z; Topcon Corp), attached to a flash-free digital camera (D7000; Nikon Corp) at 16× slit lamp optical magnification. All participants underwent Scheimpflug imaging with the Pentacam (Pentacam HR; Oculus).

Corneal Tomography and Wavefront Aberrations

Three tomographic features of subclinical edema: (1) loss of parallel isopachs; (2) displacement of the thinnest point; and (3) focal posterior depression, were graded as present or absent by 2 graders (S.P. and K.Y.T). Subclinical edema is defined as the presence of ≥2 of 3 tomographic features of subclinical edema.[26] Severity of FECD was graded separately using modified Krachmer grading.[6,27] Krachmer grade 6 eyes with clinically significant edema were excluded because the quality of corneal photography and tomography in these eyes was compromised. Wavefront aberrations of the central 6 mm optical zone of anterior and posterior corneal surfaces recorded were expressed as root mean square of the Zernike coefficient determined by the ray-tracing software, taking the refracting index of cornea as 1.3375.

Corneal Densitometry

Densitometry represents the intensity of light backscatter and is represented with values between 0 and 100 recorded in grayscale unit, where 0 indicates minimal and 100 maximal light scatter. The cornea was divided into 4 concentric zones of 0 to 2 mm, 2 to 6 mm, 6 to 10 mm, and 10 to 12 mm annuli, and its depth was automatically segmented into anterior layer (anterior 0–120 μm), central layer (between anterior 120 μm to posterior 60 μm), and posterior layer (posterior 60 μm). The average densitometry of each layer in each concentric zone was recorded. In addition, a point of highest densitometry from the posterior 20 μm of the cornea was manually segmented.

Genotyping

Genomic DNA was extracted from peripheral leukocytes of enrolled subjects using the Nucleon blood extraction kit (Amersham Biosciences). The CTG18.1 trinucleotide repeat length was determined using a previously established protocol, employing both short tandem repeat analysis and triplet-primed polymerase chain reaction.[19] Triplet-primed polymerase chain reaction served as confirmatory testing for expanded alleles (≥40 repeats) when short tandem repeat analysis was inconclusive. Subjects were subsequently categorized based on the presence or absence of expanded alleles in the CTG18.1 locus, consistent with local population data on FECD.

Statistical Analysis

All statistical analysis was performed using IBM SPSS Statistics (v23.0). Continuous data were expressed as mean ± standard deviation. Generalized estimating equation adjusted for both eyes,

estimated marginal means, was used to compare data that were not normally distributed. Multivariate regression analysis was performed to look at the association with presence of subclinical edema and CTG repeat expansion. Pearson correlation adjusted for inter-eye correlation was used to analyze the correlation between imaging parameters. P value of <0.05 was considered statistically significant.

Results

Demographics

Table 1 presents the baseline characteristics of our study population. Of the 360 eyes with FECD, 108 were included in the final analysis. The control group comprised 59 eyes from 44 healthy subjects. The majority of patients in the FECD group were female (76.9%) compared with 50.0% in the control group, and were significantly older than controls (69.3 ± 8.9 years vs. 64.4 ± 8.9 years, $P < 0.001$). Table 1 also details the distribution of FECD severity, with a mean Krachmer grading of 2.9 ± 1.2. Subclinical edema, defined as ≥2 tomographic features, was present in 49 FECD eyes (45.4%).

Distribution of Scheimpflug Imaging Parameters in FECD Eyes and Controls

Table 2 illustrates a general increase in trend in the corneal densitometry from controls, to FECD without subclinical edema, to FECD with subclinical edema, particularly within the central 0 to 2 mm and 2 to 6 mm annuli ($P < 0.01$). Higher-order aberration of the anterior and posterior cornea also shows a similar increase in trend across groups ($P < 0.001$). However, pachymetry did not differ significantly among the groups. Figure 1 shows a series of box plots representing densitometry of different corneal segmentation and annulus stratified by Krachmer grading. Within the 0 to 2 mm annulus, there was a linear increase in densitometry with higher Krachmer grades, with anterior 120 µm and posterior 60 µm showing significant increases compared with the central layer. This trend is not evident in the other corneal annuli.

Table 1. Baseline Characteristics of Eyes with Fuchs Endothelial Corneal Dystrophy and Controls

Parameter	FECD	Control	P Value
No. of subjects	83	44	
No. of eyes	108	59	
Sex, female	63 (76.9%)	22 (50%)	0.003
Age, y	69.3 ± 8.9	64.4 ± 8.9	0.004
FECD severity			
Grade 1	16 (14.8%)		
Grade 2	27 (25.0%)		
Grade 3	30 (27.8%)		
Grade 4	24 (22.2%)		
Grade 5	11 (10.2%)		
Tomographic features of subclinical edema			
Loss of parallel isopachs	53 (49.1%)		
Displacement of the thinnest point	47 (43.5%)		
Focal posterior depression	44 (40.7%)		

FECD = Fuchs endothelial corneal dystrophy.

Association of Scheimpflug Imaging Parameters and Subclinical Edema

Table 3 presents the multivariate logistic regression analysis of Scheimpflug imaging parameters associated with subclinical edema, adjusted for inter-eye correlation, gender, age, and disease severity. Corneal densitometry in the anterior 120 µm 0 to 2 mm and 2 to 6 mm annuli, central 0 to 2 mm and 2 to 6 mm annuli, and posterior 60 µm 0 to 2 mm annulus was significantly higher in eyes with subclinical edema ($P < 0.05$). The strongest association was observed in the posterior corneal HOA ($\beta = 15.068$, $P < 0.001$). Figure S2A (available at www.ophthalmologyscience.org) illustrates that posterior corneal HOA has the highest diagnostic accuracy in predicting subclinical edema (area under curve = 0.847, 95% confidence interval [CI]: 0.768−0.926). This is followed by the 0 to 2 mm annulus densitometry of the anterior 120 µm and central and posterior 60 µm of the cornea. The strength of correlation when densitometry of posterior 20 µm was used instead of posterior 60 µm was found to be comparable.

Genotyping

A subset of FECD subjects underwent genotyping analysis for the presence of ≥40 CTG trinucleotide repeats. Of 50 subjects, 22 subjects (44.0%) had positive CTG repeat expansion. Table 4 shows the multivariate regression analysis looking at the association of CTG expansion with Scheimpflug imaging parameters, adjusted for both eyes from the same subject, age, gender, and disease severity. Eyes with CTG expansion had significantly lower densitometry within the central 0 to 2 mm annulus of the anterior 120 µm layer ($\beta = -4.59$, 95% CI: −7.28 to −1.92), central layer ($\beta = -1.76$, 95% CI: −2.79 to −0.74), and posterior 60 µm layer ($\beta = -3.05$, 95% CI: −5.67 to −0.44) but thicker corneal pachymetry ($\beta = 16.87$, 95% CI: 3.81−33.83). The comparison of imaging parameters between eyes with and without CTG repeat expansion is presented in Table S5 (available at www.ophthalmologyscience.org). Among all imaging parameters, pachymetry has the highest predictive value for CTG repeat expansion (area under curve = 0.735, 95% CI: 0.607−0.862) (Fig S2B).

Discussion

Scheimpflug imaging is a versatile imaging modality for assessing FECD, providing valuable tomographic information on pachymetry, densitometry, and HOA, especially in early stages of disease where edema is not clinically evident. Our study demonstrated that densitometry and HOA were significantly higher in FECD eyes with subclinical edema, followed by those without subclinical edema, and controls. Notably, increased posterior corneal HOA was significantly associated with the presence of subclinical edema and

Table 2. Comparison of Scheimpflug Imaging Parameters between Fuchs Endothelial Corneal Dystrophy Eyes with and without Subclinical Edema and Controls

Parameters	FECD with Subclinical Edema	FECD without Subclinical Edema	Control	*P* Value
Corneal pachymetry, μm	575.4 ± 6.1	565.9 ± 5.1	561.4 ± 4.4	0.185
Corneal densitometry, GSU				
Anterior 120 μm 0–2 mm annulus	35.9 ± 1.8	27.7 ± 0.6	25.6 ± 0.7	<0.001
Anterior 120 μm 2–6 mm annulus	33.6 ± 1.5	27.4 ± 0.7	25.1 ± 0.7	<0.001
Anterior 120 μm 6–10 mm annulus	46.6 ± 2.0	43.4 ± 1.5	42.3 ± 1.7	0.232
Anterior 120 μm 10–12 mm annulus	52.0 ± 2.4	51.9 ± 1.9	52.5 ± 2.1	0.982
Central 0–2 mm annulus	18.7 ± 0.6	16.1 ± 0.3	15.8 ± 0.3	<0.001
Central 2–6 mm annulus	18.1 ± 0.6	16.6 ± 0.4	15.9 ± 0.3	0.004
Central 6–10 mm annulus	30.5 ± 1.6	30.8 ± 1.1	29.0 ± 1.2	0.516
Central 10–12 mm annulus	30.6 ± 1.1	31.3 ± 1.0	30.4 ± 0.9	0.754
Posterior 60 μm 0–2 mm annulus	21.9 ± 1.1	16.8 ± 0.6	11.4 ± 0.4	<0.001
Posterior 60 μm 2–6 mm annulus	19.2 ± 0.8	17.0 ± 0.5	11.8 ± 0.3	<0.001
Posterior 60 μm 6–10 mm annulus	26.5 ± 1.1	27.8 ± 0.8	21.3 ± 0.7	<0.001
Posterior 60 μm 10–12 mm annulus	26.4 ± 1.0	28.8 ± 0.8	25.9 ± 0.7	0.012
Anterior corneal HOA, μm	0.79 ± 0.04	0.64 ± 0.03	0.54 ± 0.02	<0.001
Posterior corneal HOA, μm	0.40 ± 0.03	0.22 ± 0.01	0.19 ± 0.01	<0.001

FECD = Fuchs endothelial corneal dystrophy; GSU = grayscale unit; HOA = higher-order aberration.

demonstrated the highest diagnostic accuracy among all parameters, underscoring its potential as an early marker of corneal decompensation. Additionally, we identified a novel association between CTG trinucleotide repeat expansion and corneal densitometry within the central 0 to 2 mm annulus. Eyes with CTG expansion exhibited significantly lower densitometry values compared with nonexpansion eyes, suggesting a distinct morphologic phenotype.

Densitometry measures the amount of light backscatter from the cornea, and age-related loss of corneal transparency can lead to increased densitometry even in normal eyes.[28] In FECD, the presence of guttae and diseased Descemet membrane exacerbates light backscatter.[14,15] Our study found that the mean densitometry values across all corneal segmentations were significantly higher in FECD eyes compared with controls. Notably, within the central 6 mm annulus, the posterior cornea was disproportionately affected, with posterior 60 μm densitometry values approximately double those of controls. Similarly, the posterior 60 μm 0 and 2 mm annulus densitometry showed the greatest difference between severity grades, nearly doubling from grade 1 to grade 5 (Table S6, available at

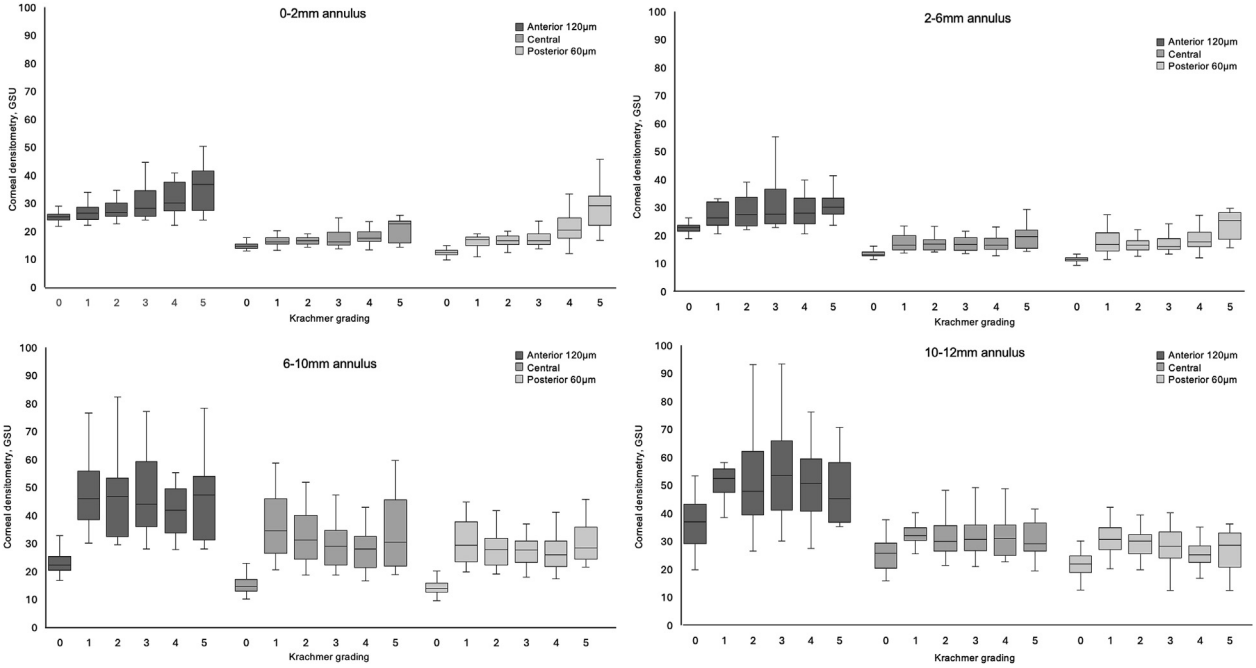


Figure 1. Distribution of corneal densitometry of controls and Fuchs endothelial corneal dystrophy with increasing severity stratified by 4 concentric annuli. GSU = grayscale unit.

Table 3. Multivariate Logistic Regression Analysis of Scheimpflug Imaging Parameters Associated with Subclinical Edema

Parameters	β (95% CI)	*P* Value
Corneal pachymetry	−0.002 (−0.013 to 0.009)	0.726
Corneal densitometry		
Anterior 120 μm 0–2 mm annulus	0.139 (0.053 to 0.224)	0.002
Anterior 120 μm 2–6 mm annulus	0.110 (0.045 to 0.176)	0.001
Central 0–2 mm annulus	0.313 (0.087 to 0.539)	0.007
Central 2–6 mm annulus	0.125 (0.018 to 0.232)	0.022
Posterior 60 μm 0–2 mm annulus	0.126 (0.007 to 0.246)	0.037
Posterior 60 μm 2–6 mm annulus	0.054 (−0.034 to 0.143)	0.226
Anterior cornea HOA	1.019 (−0.762 to 2.800)	0.262
Posterior cornea HOA	15.068 (7.546 to 22.590)	<0.001

CI = confidence interval; HOA = higher-order aberration.

www.ophthalmologyscience.org). These findings are consistent with previous studies demonstrating a progressive increase in light backscatter from both the anterior and posterior layers with advancing disease severity.[14,15,29] The middle stromal layer, composed of a highly organized extracellular matrix and collagen fibrils, typically exhibits lower light scatter.[30] On a vertical densitogram display, the reduced densitometry of the middle layer is seen as a dip flanked by 2 peaks of higher densitometry from the anterior and posterior layers, often referred to as the "hammock sign."[14] Interestingly, our results showed that the central densitometry showed a corresponding increase with increasing disease severity (Fig 1), suggesting that stromal involvement may occur early in FECD and contribute to visual degradation.[31]

Subclinical edema indicates early corneal decompensation in FECD. A revised classification using Scheimpflug stratifies FECD into 3 categories—clinically evident edema, subclinical edema, and no edema—in an attempt to better identify patients who may benefit from early surgical intervention.[10,11] Given the current shift toward earlier endothelial keratoplasty, recognizing eyes with subclinical edema or early signs of visual degradation is increasingly important.[32] Tomographic features indicative of subclinical edema, such as loss of parallel isopachs, displacement of the thinnest point, and focal posterior corneal surface depression, have been shown to predict disease progression and the likelihood of requiring future intervention.[11] In addition, subclinical edema has been associated with reduced contrast sensitivity, further highlighting its clinical relevance.[26] Our study demonstrated that eyes with subclinical edema exhibited significantly higher densitometry and HOA, with posterior corneal HOA demonstrating the strongest association. These findings support the hypothesis that increased light backscatter and optical aberrations from the posterior cornea contribute substantially to the decline in visual function in early FECD.

Table 4. Multivariate Regression Analysis of Scheimpflug Imaging Parameters Associated with CTG Repeat Expansion

Parameters	β (95% CI)	*P* Value
Corneal pachymetry	16.87 (3.81–33.83)	0.051
Corneal densitometry		
Anterior 120 μm, 0–2 mm annulus	−4.59 (−7.28 to −1.92)	<0.001
Anterior 120 μm, 2–6 mm annulus	−2.74 (−5.62 to 0.14)	0.062
Central cornea, 0–2 mm annulus	−1.76 (−2.79 to −0.74)	<0.001
Central cornea, 2–6 mm annulus	−0.25 (−1.79 to 1.47)	0.777
Posterior 60 μm, 0–2 mm annulus	−3.05 (−5.67 to −0.44)	0.002
Posterior 60 μm, 2–6 mm annulus	−0.92 (−3.04 to 1.21)	0.397
Anterior corneal HOA	0.02 (−0.08 to 0.13)	0.650
Posterior corneal HOA	−0.02 (−0.14 to 0.002)	0.058

CI = confidence interval; HOA = higher-order aberration.

Posterior cornea densitometry has also been shown to correlate closely with reduced visual function, highlighting the potential impact of light backscatter on vision.[17] The increased fibrosis, thickening of Descemet membrane, and elevated keratocyte density in the pre-Descemet layer contribute to increased posterior light scatter.[33] To investigate whether guttae was the primary cause of aberrations, we further segmented a thinner layer of just the posterior 20 μm to approximate the thickness of the endothelial layer. We had theorized that a thinner segmentation would correlate better with disease severity but found no difference when comparing the posterior 20 and 60 μm 0 to 2 mm annulus densitometry. Given that a thinner posterior segmentation did not improve the specificity and correlation with disease, we conclude that presegmented posterior 60 μm remains a reliable and practical parameter in assessing FECD.

We also examined the association between Scheimpflug parameters with presence of CTG repeat expansion given recent evidence suggesting that it may be pathogenic in FECD and linked to a greater risk of disease progression.[24] An illustration of CTG repeat positive versus negative eyes with varying disease severity is shown in Figure 3. Our previous work suggests that subjects with expanded CTG allele had more severe disease and were twice as likely to develop corneal decompensation at 5 years.[19,24] It was also reported that in severe FECD, eyes with positive repeat expansion demonstrated higher corneal densitometry in the posterior cornea.[29] This contradicts our findings, where, despite a small sample size, we found that repeat expansion positive eyes had significantly lower densitometry. The involvement of the TCF4 gene in corneal endothelial cellular migration may have influenced the clarity of the cornea in early disease and could play a role in this distinct phenotype.[25] However, we acknowledge that the exclusion of eyes with scarring and late corneal decompensation may have influenced the observed association. Future studies could explore architectural differences between repeat expansion positive and negative eyes using more detailed imaging, such as high-resolution OCT in vivo.

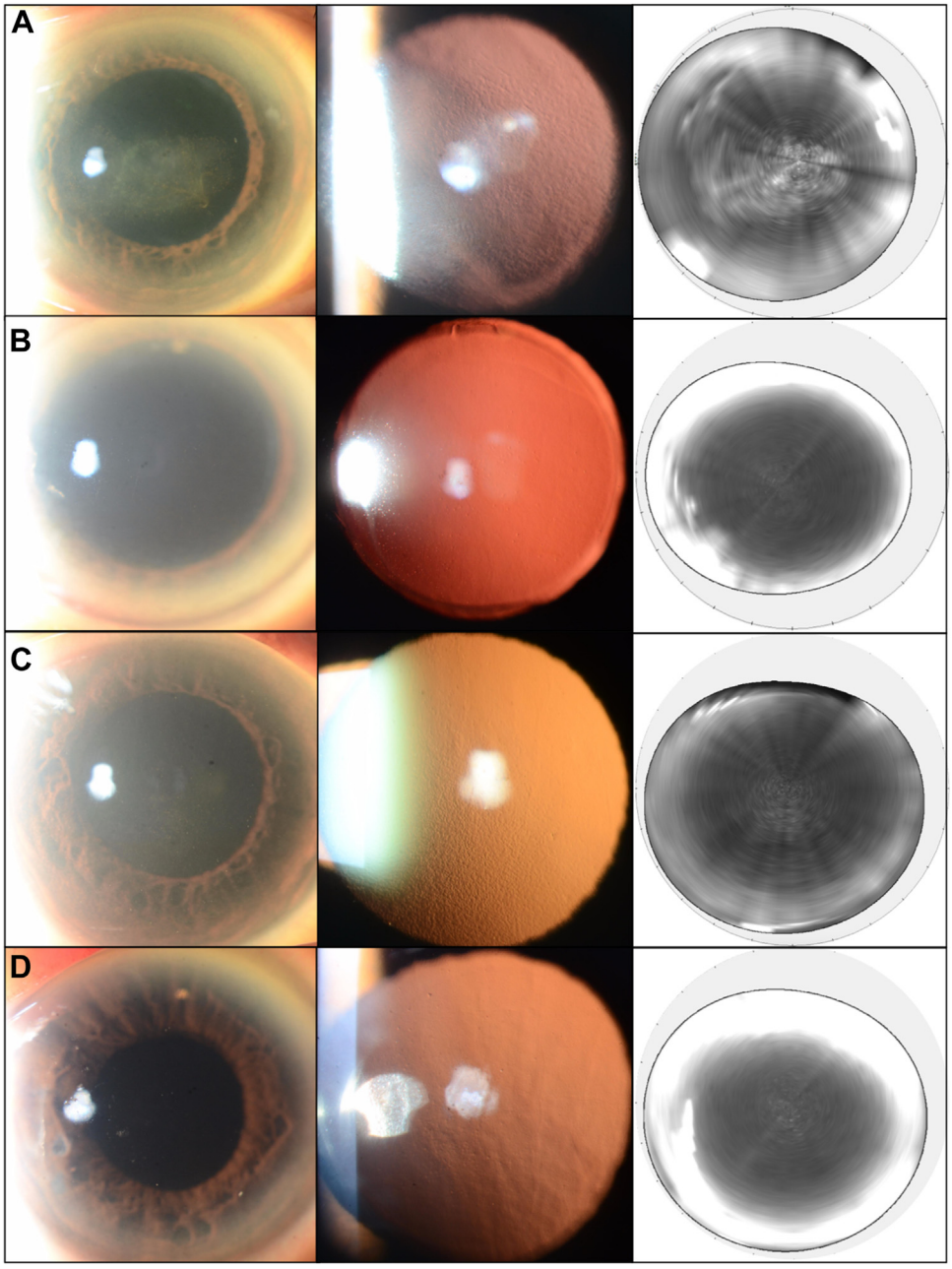


Figure 3. A series of 4 eyes, each shown with a corresponding sclerotic scatter photo, retroillumination photo, and densitometry map. Row **A** features a CTG repeat positive eye with grade 5 disease demonstrating much brighter signals on the densitometry map compared with that in row **B,** similarly a CTG repeat positive eye with grade 1 disease. The last 2 rows illustrate CTG repeat negative eyes, with row **C** illustrating a grade 5 disease, and row **D** illustrating a grade 1 disease.

This study has several limitations primarily due to its retrospective and cross-sectional design and the absence of direct comparisons with visual function outcomes. Visual functions can be influenced by various ocular comorbidities, including cataracts, macular pathology, and optic nerve disease, which may confound its correlation with corneal imaging parameters. Therefore, the independent associations of optical aberration and densitometry with subclinical edema demonstrated in our study provide a valuable indicator of progression without relying on subjective parameters. These parameters could facilitate decisions on surgical intervention independent of cataract status and visual acuity. Longitudinal studies are warranted to assess the progression of subclinical edema and elucidate the temporal relationship between morphologic changes and clinical outcomes. Additionally, genetic analysis was available for only half of the FECD cohort. Although this subset is representative of our local population and may be

generalizable to other Asian populations, further studies on Western populations, where expansion rates are higher, are needed to validate these findings across different genetic backgrounds.

Conclusion

Scheimpflug imaging is a valuable tool in the assessment of FECD. Eyes with tomographic features of subclinical edema are associated with higher densitometry and posterior corneal HOA. We also found a novel correlation between the presence of CTG repeat expansion with lower corneal densitometry, which could indicate differences in genetic phenotypes. The interpretation of multiple parameters, including densitometry, HOA analysis, and tomographic features indicative of subclinical edema, may offer a more comprehensive approach to monitoring disease progression, especially in the assessment of early FECD.

Footnotes and Disclosures

Originally received: November 21, 2024.
Final revision: May 14, 2025.
Accepted: May 14, 2025.
Available online: May 20, 2025. Manuscript no.: XOPS-D-24-00510R2.

[1] Singapore National Eye Centre, Singapore.

[2] Ophthalmology and Visual Sciences Academic Clinical Program, Duke-NUS, Singapore.

[3] Tissue Engineering and Cell Therapy Research Group, Singapore Eye Research Institute, Singapore.

[4] Singapore Eye Research Institute, Singapore.

[5] Woodlands Health Department of Ophthalmology, Singapore.

[6] Department of Ophthalmology, University of Texas Southwestern Medical Center, Dallas, Texas.

Presented at the American Academy of Ophthalmology Annual Meeting, San Francisco, California, November 3-6, 2023.

Disclosure(s):

The Article Publishing Charge (APC) for this article was paid by Singhealth Fund-SNEC (SHF-SNEC).

All authors have completed and submitted the ICMJE disclosures form.

The authors made the following disclosures:

J.S.M.: The Molecular Therapeutics Programme, H17/01/a0/012, Industry Alignment Fund (Agency of Science, Technology and Research, Health and Biomedical Sciences (HBMS) Singapore); The Molecular Therapeutics Programme, R1509/2018, SingHealth Fund. V.V.M.: Grants or contracts from any entity − National Institutes of Health, Bethesda, Maryland (grants R01EY022161 [V.V.M.] and P30EY030413 [V.V.M.]); Challenge Grant, Research to Prevent Blindness, New York.

The other authors have no proprietary or commercial interest in any materials discussed in this article.

HUMAN SUBJECTS: Human subjects were included in this study. SingHealth Centralized Institutional Review Board approved the study. All research adhered to the tenets of the Declaration of Helsinki. Due to the retrospective design and use of anonymized data, the requirement for patient consent was waived.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Poh, Soh, Mehta

Data collection: Poh, Tey, Neo, Lee, Mehta

Analysis and interpretation: Poh, Peh, Htoon, Mootha, Metha

Obtained funding: Mehta

Overall responsibility: Poh, Metha

Abbreviations and Acronyms:
CI = confidence interval; **FECD** = Fuchs endothelial corneal dystrophy; **HOA** = higher-order aberration.

Keywords:
Fuchs endothelial corneal dystrophy, Scheimpflug imaging, Densitometry, Optical aberrations, Genetics.

Correspondence:
Jodhbir S. Mehta, PhD, Singapore Eye Research Institute, 20 College Road, The Academia, Discovery Tower Level 6, Singapore 169856. E-mail: jodhbir.s.mehta@singhealth.com.sg.

References

1. Wilson SE, Bourne WM. Fuchs' dystrophy. *Cornea*. 1988;7:2−18.
2. Louttit MD, Kopplin LJ, Igo RP, et al. A multicenter study to map genes for Fuchs endothelial corneal dystrophy: baseline characteristics and heritability. *Cornea*. 2012;31:26−35.
3. Blitzer AL, Colby KA. Update on the surgical management of Fuchs endothelial corneal dystrophy. *Ophthalmol Ther*. 2020;9:757−765.
4. Woo JH, Ang M, Htoon HM, Tan D. Descemet membrane endothelial keratoplasty versus descemet stripping automated endothelial keratoplasty and penetrating keratoplasty. *Am J Ophthalmol*. 2019;207:288−303.
5. Singh A, Zarei-Ghanavati M, Avadhanam V, Liu C. Systematic review and meta-analysis of clinical outcomes of descemet membrane endothelial keratoplasty versus descemet stripping endothelial keratoplasty/descemet stripping automated endothelial keratoplasty. *Cornea*. 2017;36:1437−1443.
6. Krachmer JH, Purcell JJ, Young CW, Bucher KD. Corneal endothelial dystrophy: a study of 64 families. *Arch Ophthalmol*. 1978;96:2036−2039.
7. Repp DJ, Hodge DO, Baratz KH, et al. Fuchs' endothelial corneal dystrophy: subjective grading versus objective grading based on the central-to-peripheral thickness ratio. *Ophthalmology*. 2013;120:687−694.
8. Matthaei M, Hribek A, Clahsen T, et al. Fuchs endothelial corneal dystrophy: clinical, genetic, pathophysiologic, and therapeutic aspects. *Annu Rev Vis Sci*. 2019;5:151−175.
9. Seitzman GD, Gottsch JD, Stark WJ. Cataract surgery in patients with Fuchs' corneal dystrophy: expanding recommendations for cataract surgery without simultaneous keratoplasty. *Ophthalmology*. 2005;112:441−446.
10. Sun SY, Wacker K, Baratz KH, Patel SV. Determining subclinical edema in fuchs endothelial corneal dystrophy: revised

classification using scheimpflug tomography for preoperative assessment. *Ophthalmology*. 2019;126:195–204.
11. Patel SV, Hodge DO, Treichel EJ, et al. Predicting the prognosis of fuchs endothelial corneal dystrophy by using scheimpflug tomography. *Ophthalmology*. 2020;127:315–323.
12. Ishikawa S, Kato N, Takeuchi M. Quantitative evaluation of corneal epithelial edema after cataract surgery using corneal densitometry: a prospective study. *BMC Ophthalmol*. 2018;18:334.
13. Cennamo G, Forte R, Aufiero B, La Rana A. Computerized Scheimpflug densitometry as a measure of corneal optical density after excimer laser refractive surgery in myopic eyes. *J Cataract Refract Surg*. 2011;37:1502–1506.
14. Chu HY, Hsiao CH, Chen PYF, et al. Corneal backscatters as an objective index for assessing fuchs' endothelial corneal dystrophy: a pilot study. *J Ophthalmol*. 2017;2017:8747013.
15. Wacker K, McLaren JW, Amin SR, et al. Corneal high-order aberrations and backscatter in fuchs' endothelial corneal dystrophy. *Ophthalmology*. 2015;122:1645–1652.
16. Shah K, Eghrari AO, Vanner EA, et al. Scheimpflug corneal densitometry values and severity of guttae in relation to visual acuity in fuchs endothelial corneal dystrophy. *Cornea*. 2022;41:692–698.
17. Kai C, Oie Y, Nishida N, et al. Associations between visual functions and severity gradings, corneal scatter, or higher-order aberrations in fuchs endothelial corneal dystrophy. *Invest Ophthalmol Vis Sci*. 2024;65:15.
18. Kuot A, Hewitt AW, Snibson GR, et al. TGC repeat expansion in the TCF4 gene increases the risk of Fuchs' endothelial corneal dystrophy in Australian cases. *PloS One*. 2017;12:e0183719.
19. Soliman AZ, Xing C, Radwan SH, et al. Correlation of severity of fuchs endothelial corneal dystrophy with triplet repeat expansion in TCF4. *JAMA Ophthalmol*. 2015;133:1386–1391.
20. Luther M, Grünauer-Kloevekorn C, Weidle E, et al. TGC-repeats im intron 2 des TCF4-gens haben eine große Vorhersagekraft bezüglich Fuchs-Hornhautendotheldystrophie. *Klin Monatsblätter Für Augenheilkd*. 2016;233:187–194.
21. Xing C, Gong X, Hussain I, et al. Transethnic replication of association of CTG18.1 repeat expansion of TCF4 gene with Fuchs' corneal dystrophy in chinese implies common causal variant. *Invest Ophthalmol Vis Sci*. 2014;55:7073–7078.
22. Nakano M, Okumura N, Nakagawa H, et al. Trinucleotide repeat expansion in the TCF4 gene in fuchs' endothelial corneal dystrophy in Japanese. *Invest Ophthalmol Vis Sci*. 2015;56:4865–4869.
23. Maeno S, Oie Y, Koto R, et al. Comparison of Scheimpflug and anterior segment optical coherence tomography imaging parameters for Japanese patients with fuchs endothelial corneal dystrophy with and without TCF4 repeat expansions. *Cornea*. 2024;43:805–811.
24. Soh YQ, Peh Swee Lim G, Htoon HM, et al. Trinucleotide repeat expansion length as a predictor of the clinical progression of Fuchs' endothelial corneal dystrophy. *PLoS One*. 2019;14:e0210996.
25. Yan J, Mehta S, Patel K, et al. Transcription factor 4 promotes increased corneal endothelial cellular migration by altering microtubules in Fuchs endothelial corneal dystrophy. *Sci Rep*. 2024;14:10276.
26. Friedrich M, Hofmann CA, Chychko L, et al. Influence of subclinical corneal edema on contrast sensitivity in Fuchs endothelial corneal dystrophy. *Cornea*. 2024;43:1154–1161.
27. Eghrari AO, Garrett BS, Mumtaz AA, et al. Retroillumination photography analysis enhances clinical definition of severe Fuchs corneal dystrophy. *Cornea*. 2015;34:1623–1626.
28. Wu Z, Wang Y, Zhang L, et al. Distribution of corneal densitometry and its correlation with ocular stray light in healthy eyes. *Zhonghua Yan Ke Za Zhi*. 2014;50:20–26.
29. Gillings M, Mastro A, Zhang X, et al. Loss of corneal nerves and corneal haze in patients with Fuchs' endothelial corneal dystrophy with the transcription factor 4 gene trinucleotide repeat expansion. *Ophthalmol Sci*. 2023;3:100214.
30. Cankaya AB, Tekin K, Kiziltoprak H, et al. Assessment of corneal backward light scattering in the healthy cornea and factors affecting corneal transparency. *Jpn J Ophthalmol*. 2018;62:335–341.
31. Amin SR, Baratz KH, McLaren JW, Patel SV. Corneal abnormalities early in the course of Fuchs' endothelial dystrophy. *Ophthalmology*. 2014;121:2325–2333.
32. Patel SV. Towards clinical trials in fuchs endothelial corneal dystrophy: classification and outcome measures—The Bowman club lecture 2019. *BMJ Open Ophthalmol*. 2019;4:e000321.
33. Schrems-Hoesl LM, Schrems WA, Cruzat A, et al. Cellular and subbasal nerve alterations in early stage Fuchs' endothelial corneal dystrophy: an in vivo confocal microscopy study. *Eye*. 2013;27:42–49.